Do demand shifts matter for forecasting and stock?

Walmart M5, California (726 high-volume items, sales summed across stores), forecast evaluation 2015-12-07 to 2016-05-22, replicated in Texas and Wisconsin. M5 has no COVID-era data. Generated from outputs/results_facts.json.

Do demand shifts exist, and can we detect them?

Partly STRONG EVIDENCE

An estimated 63.5% of items have a real shift (95% CI 59.8% to 67.2%), after removing chance alarms.

Caveat: The detector is coarse: a sudden 20% shift is caught only 18% of the time.

Which forecast is best?

LightGBM MODERATE EVIDENCE

LightGBM: median error 0.744 vs 0.976 for the simple baseline (MASE; lower is better).

Caveat: It under-forecasts by 3.6% in total.

Does retraining after a shift help?

No STRONG EVIDENCE

None of 4 retraining methods improved forecasts. Simply restarting from the shift made error slightly worse (+0.0033 MASE, n = 488 items), and the same holds in 2 other states.

Caveat: Tested with one model (LightGBM) and four simple methods; small shifts are invisible to the detector.

Does a better forecast cut stock cost?

Yes STRONG EVIDENCE

At a 95% target, LightGBM-based reorder points hold 16.6% less stock at 10.1% lower cost than the simple baseline; clearly lower in 3 of 3 time windows.

Caveat: Lead time and costs are assumptions, not measured data.

Key numbers

Items modelled	726 high-volume California items (23.8% of items, 66.4% of units)
False alarms (seasonal null)	CUSUM 11.9%, PELT 16.1%
Items with a real shift	63.5% (95% CI 59.8% to 67.2%), n = 718
High-confidence breaks	171 of 488
Median forecast error (MASE)	LightGBM 0.744, ETS 0.755, seasonal naive 0.976
Retraining from the shift	+0.0033 MASE (95% CI +0.0015 to +0.0053), n = 488
Calibrated 90% band coverage	89.9% (was 85.5%)
Static LightGBM vs naive, 95% target ASSUMED COSTS	16.6% less inventory, 10.1% lower cost
Calibrated safety stock, 95% target ASSUMED COSTS	total cost -31.1%

Replication (Texas and Wisconsin, every setting frozen at the California value)

State	Conclusion	Result	Evidence
TX	R1 breaks are common	holds	CUSUM break rate 0.6635
TX	R2 LightGBM beats seasonal naive	holds	median MASE 0.7342 vs 0.9479, p = 6.76e-72
TX	R3 LightGBM at least as good as ETS	holds	median MASE 0.7342 vs ETS 0.7393
TX	R4 truncate-adaptation does not help	holds	median diff +0.0046 (CI +0.0022 to +0.0077), p = 0.0000, n = 353
TX	R5 static LightGBM cheaper than naive (95%)	holds	47727 vs 56254
TX	R6 adaptive not cheaper than static (95%)	holds	total diff +2234, 95% CI +252 to +4285
TX	R7 99% cheapest of 90/95/99	holds	cheapest = 0.99
WI	R1 breaks are common	holds	CUSUM break rate 0.6488
WI	R2 LightGBM beats seasonal naive	holds	median MASE 0.7786 vs 1.0252, p = 4.86e-88
WI	R3 LightGBM at least as good as ETS	holds	median MASE 0.7786 vs ETS 0.8049
WI	R4 truncate-adaptation does not help	holds	median diff +0.0071 (CI +0.0052 to +0.0096), p = 0.0000, n = 386

WI	R5 static LightGBM cheaper than naive (95%)	holds	54896 vs 60474
WI	R6 adaptive not cheaper than static (95%)	holds	total diff +168, 95% CI -2393 to +2460
WI	R7 99% cheapest of 90/95/99	holds	cheapest = 0.99

Lead time, holding and stockout costs are assumptions, not observed data. Full results: docs/RESULTS.md. Interactive version: dashboard/app.py.